a

Cohort fine-tuning

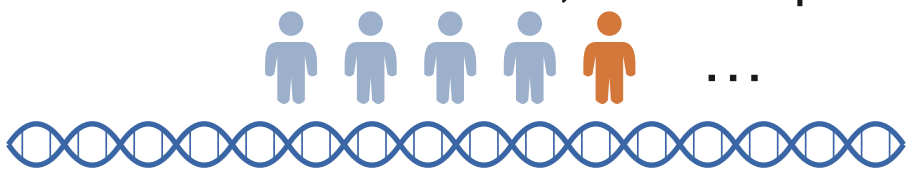
8 released checkpoints / 4 families

NT **DNABERT**

DNABERT-2 **HyenaDNA**

Human-derived chr22 records

Rare variants: ≤5 / 2,548 samples



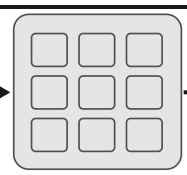
96 nt record



Nominal prefix **48 nt target**

1 / 4 / 16 copies · random offsets

Fine-tune



White-box audit **Shared weights**

b

Two tests of the same weights

DNABERT-2 · chr22 · final checkpoint · 3 seeds

Sequence reproduction



Prefix Hidden target

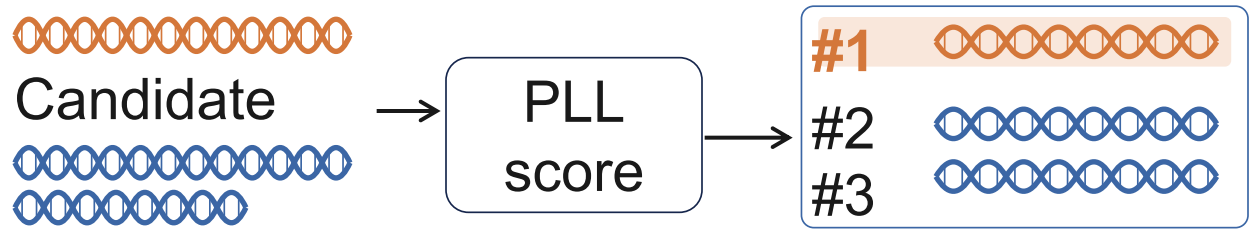
Prefix: suffix masked
Search: flanks visible

Copies	Prefix	BPE search*
1 copy	**0%**	**0%**
16 copies	**40–44%**	**0%**

*Own window; historical BPE scorer.

Candidate membership

Single-copy candidate vs 100 alternatives



92–96%

Own (privileged)

18–26%

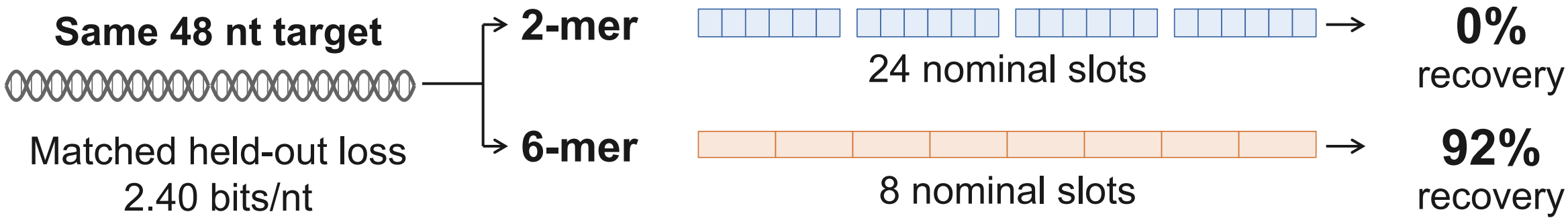
Fresh window

Controls: own matched; fresh independently drawn (final 0/50). Chance: 0.99%.

c

Controlled tokenization: recovery and conditional prediction

Fixed non-embedding body, objective and corpus; 16-copy synthetic canaries



End-of-training target prediction and recovery

Covering-token surprise (bits) / 48 target nt; includes trailing host letters

Tokenizer	Planted	Control − planted	Recovery
2-mer	0.824	2.44	0%
6-mer	0.0052	2.79	100%